

Reactive Phosphonate Species | *Very Important Paper* |

**Solid-State Structure of a Bromo(difluoromethylenephosphonato)-cuprate – a Key Intermediate for the Synthesis of  $\alpha,\alpha$ -Difluoromethylenephosphonates**

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*Dedicated to the memory of Professor D. J. Burton*

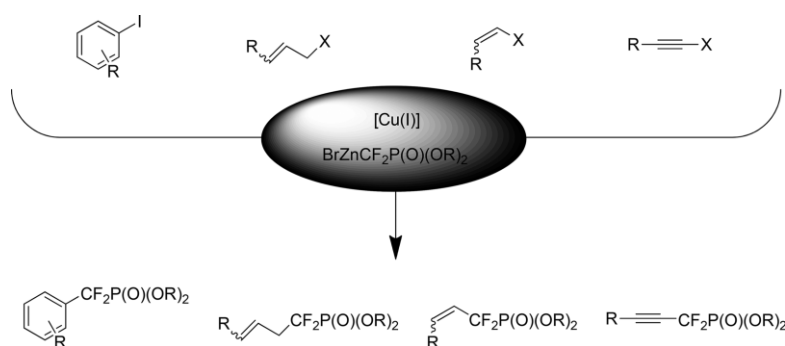
**Abstract:** Many approaches to introduce the  $\text{CF}_2\text{P}(\text{O})(\text{OR})_2$  unit into organic molecules make use of  $\text{Cu}^{\text{I}}$ -catalyzed couplings between organic halides and  $\text{BrZnCF}_2\text{P}(\text{O})(\text{OR})_2$ . Despite the great importance of this process for the synthesis of  $\alpha,\alpha$ -difluoromethylenephosphonates, the constitution and molecular structures of the organozinc precursor and the catalytically relevant copper/zinc complex are still unknown. Here we report on the

isolation of these two species, their spectroscopic characterization, and the determination of their structure by single-crystal X-ray diffraction. First orienting experiments on the stability of the cuprate reagent in solution at room temperature as well as its reactivity towards allyl bromides were subsequently performed.

## Introduction

Fluorine or fluorine-containing substituents influence the biological properties of organic molecules such as their bioavailability or metabolic stability.<sup>[1]</sup> One prominent functionality in such species is the difluoromethylenephosphonate unit  $\text{CF}_2\text{P}(\text{O})(\text{OR})_2$ . As  $\alpha,\alpha$ -difluoromethylenephosphonates can be considered stable metabolic phosphate mimics,<sup>[2]</sup> they are challenging candidates for the design of novel pharmaceuticals.

Aryldifluoromethylenephosphonates are potent inhibitors of tyrosine phosphatase, which is an important enzyme for cellular signal transduction.<sup>[3]</sup> The  $\text{CF}_2\text{P}(\text{O})(\text{OR})_2$  unit is available by fluorination of keto- or methylphosphonates with diethylaminosulfurtrifluoride (DAST).<sup>[4]</sup> Aggressive fluorinating agents may be circumvented by methods based upon metalated  $\text{CF}_2\text{P}(\text{O})(\text{OR})_2$  derivatives. One prominent synthetic approach to  $\alpha,\alpha$ -difluoromethylenephosphonates is centered around  $\text{BrZnCF}_2\text{P}(\text{O})(\text{OR})_2$ .



Scheme 1. Coupling reactions with  $\text{BrZnCF}_2\text{P}(\text{O})(\text{OR})_2$  in the presence of  $\text{Cu}^{\text{I}}$  species.

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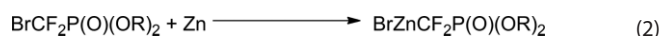
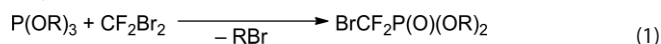
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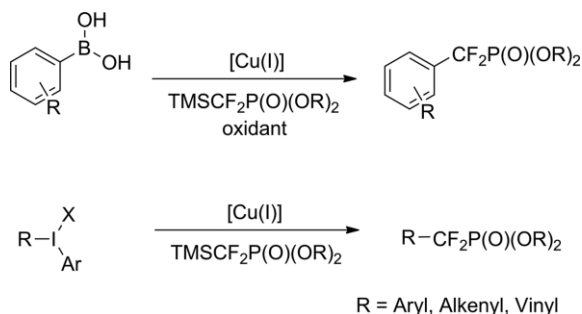
Supporting information for this article is available on the WWW under <https://doi.org/10.1002/ejic.201700526>.

Treatment of bromodifluoromethyldialkylphosphonates with zinc leads to the formation of  $\text{BrZnCF}_2\text{P}(\text{O})(\text{OR})_2$  [Equation (2)]. The required bromodifluoromethyldialkylphosphonates are easily accessible by the Michaelis–Arbuzov reaction of trialkylphosphites with  $\text{CF}_2\text{Br}_2$  [Equation (1)].<sup>[5]</sup> The organozinc compound reacts readily with various acylation agents.<sup>[6]</sup> In the presence

of Cu<sup>I</sup> salts the CF<sub>2</sub>P(O)(OR)<sub>2</sub> unit can also be coupled to allyl halides,<sup>[7a]</sup> aryl iodides,<sup>[7b–7c,7j]</sup> alkenyl halides,<sup>[7d–7h]</sup> and haloalkynes<sup>[7i]</sup> (Scheme 1).

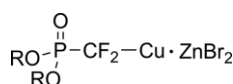


In addition to organozinc chemistry, several other copper-mediated couplings such as those of boronic acids or organoiodine(III) halides with Me<sub>3</sub>SiCF<sub>2</sub>P(O)(OR)<sub>2</sub> have been reported.<sup>[8]</sup> (Scheme 2).



Scheme 2. Copper(I) mediated syntheses of difluoromethylenephosphonates.

Despite the number of reports on copper mediated couplings with BrZnCF<sub>2</sub>P(O)(OR)<sub>2</sub>, the constitution and molecular structure of the copper reagent remained virtually obscure. Yokomatsu and Burton<sup>[9,7e]</sup> postulated a mixed Cu/Zn species as the reactive intermediate (Scheme 3).

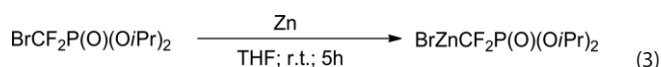


Scheme 3. Postulated structure of the reactive organocopper intermediate.<sup>[9,7e]</sup>

As yet, neither the organocopper intermediate nor the organozinc precursor have been isolated and structurally authenticated by X-ray diffraction or by means of spectroscopy. In the course of a program on difluoroorganic compounds, we were interested in the structure/reactivity properties of the catalytic system under discussion.

## Results and Discussion

Following Burton's protocol we synthesized BrZnCF<sub>2</sub>P(O)(OiPr)<sub>2</sub> by treatment of BrCF<sub>2</sub>P(O)(OiPr)<sub>2</sub> with zinc metal and confirmed the relevant spectroscopic data as given by the authors [Equation (3)].<sup>[6]</sup>



Crystallization of [Zn(Br){CF<sub>2</sub>P(O)(OiPr)<sub>2</sub>}(thf)]<sub>2</sub> from a THF solution at –28 °C afforded colorless needles. The air-sensitive crystals were stored under a N<sub>2</sub> atmosphere at room temperature for months without any decomposition and were subjected to an X-ray diffraction study.

The compound crystallizes in the triclinic space group *P*1̄ (Figure 1). The analysis reveals two crystallographically independent molecules, both BrZnCF<sub>2</sub>P(O)(OiPr)<sub>2</sub> units are arranged at a center of inversion in an antiparallel disposition. The zinc atoms are at the center of a distorted tetrahedron [C1–Zn1–Br1 126.3(1)° and C12–Zn2–Br2 126.8(1)°] which is completed by a THF ligand and the P=O group of the adjacent molecule [O1'–Zn1–O4 93.2(1)° and O5–Zn2–O8 93.5(1)°]. The C–Zn distances are 203.4(2) and 202.7(2) pm, respectively.

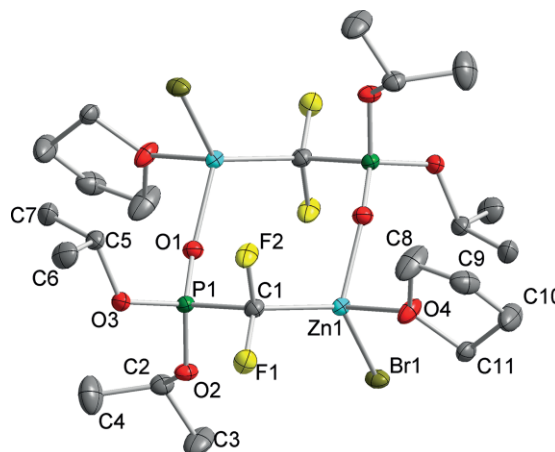
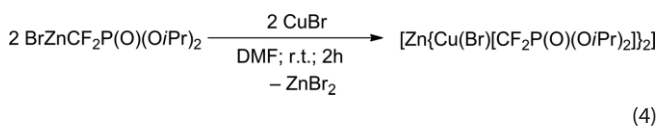


Figure 1. Molecular structure of [Zn(Br){CF<sub>2</sub>P(O)(OiPr)<sub>2</sub>}(thf)]<sub>2</sub> in the solid state (thermal ellipsoids are set to 50 % probability, hydrogen atoms and the second molecule are omitted for clarity).

In the IR spectrum (Figure 3), the band with the highest intensity at 1015 cm<sup>–1</sup> results from the P–O–C stretching vibration. The P=O stretching vibration is assigned to the intense band at 1179 cm<sup>–1</sup>. The vibrational stretching of the CF<sub>2</sub> unit gives rise to a band at 1099 cm<sup>–1</sup>.

In accordance with the suggestion of Burton and Yokomatsu,<sup>[9,7e]</sup> the addition of one equivalent of CuBr to a DMF solution of BrZnCF<sub>2</sub>P(O)(OiPr)<sub>2</sub> gives rise to the formation of the heteroorganocuprate [Zn{Cu(Br){CF<sub>2</sub>P(O)(OiPr)<sub>2</sub>}}<sub>2</sub>(dmf)<sub>4</sub>] [Equation (4)], which was isolated from a cold THF solution as colorless needles. The light- and air-sensitive cuprate can be stored under a N<sub>2</sub> atmosphere at –28 °C in the dark for weeks.



An X-ray structural study shows the picture of a heteroorganocuprate structure (Figure 2). The compound crystallizes in the monoclinic space group *P*2<sub>1</sub>/*c* with Zn at the center of inversion. Two cuprate units are coordinated to the central Zn cation by their P=O functionality [O1–Zn1 206.2(1) pm]. Four equatorial positions at the Zn atom are occupied by the oxygen atoms of four DMF ligands. Within the structure, the copper atom adopts an almost linear arrangement [C1–Cu1–Br1 178.0(1)°] with a C1–Cu1 bond length of 192.0(2) pm. In the IR spectrum of the product, bands at 1109 and 1053 cm<sup>–1</sup> were assigned to the symmetric and antisymmetric ν(CF) vibrations (Figure 3). The very intense band at 1643 cm<sup>–1</sup> is due to the

C=O stretching of the coordinated dimethylformamide. Intense bands at  $1376\text{ cm}^{-1}$  (C–N stretching) and at  $682\text{ cm}^{-1}$  (C=O deformation) are assigned to the DMF ligand.

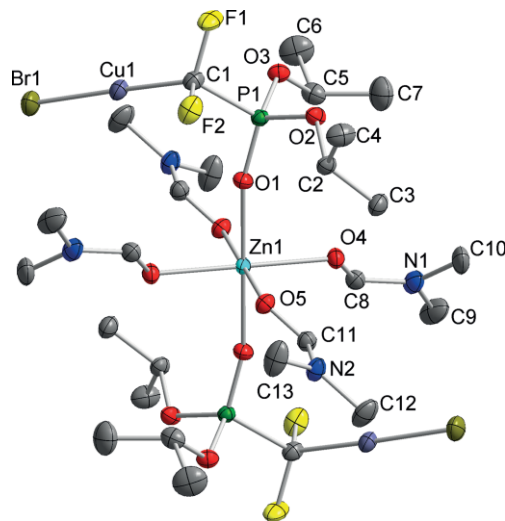


Figure 2. Molecular structure of  $[\text{Zn}\{\text{Cu}(\text{Br})\{\text{CF}_2\text{P}(\text{O})(\text{OiPr})_2\}_2\}(\text{dmf})_4]$  in the solid state (thermal ellipsoids are set to 50 % probability, hydrogen atoms omitted).

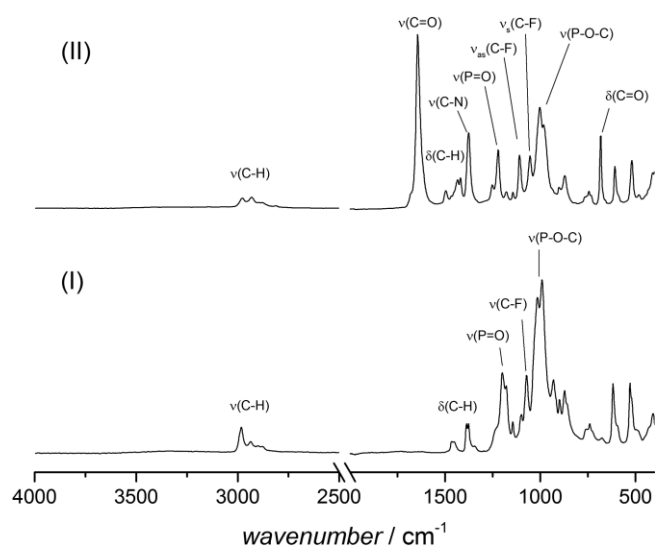


Figure 3. IR spectra of  $[\text{Zn}(\text{Br})\{\text{CF}_2\text{P}(\text{O})(\text{OiPr})_2\}(\text{thf})_2]$  (ATR, I) and  $[\text{Zn}\{\text{Cu}(\text{Br})\{\text{CF}_2\text{P}(\text{O})(\text{OiPr})_2\}_2\}(\text{dmf})_4]$  (ATR, II).

The  $^{19}\text{F}$  NMR spectrum of  $[\text{Zn}\{\text{Cu}(\text{Br})\{\text{CF}_2\text{P}(\text{O})(\text{OiPr})_2\}_2\}(\text{dmf})_4]$  in  $[\text{D}_6]\text{DMSO}$  reveals a doublet at  $-114.0\text{ ppm}$  ( $^2J_{\text{PF}} = 91\text{ Hz}$ ). In the  $^{31}\text{P}$  NMR spectrum, the phosphorus atom resonates at  $\delta = 13.7\text{ ppm}$  as a triplet of triplets with a  $^2J_{\text{PF}}$  coupling of  $91\text{ Hz}$ . Whereas the organocuprate **1** is insoluble in ethereal solvents, its solutions in  $\text{CH}_3\text{CN}$ , DMF, and DMSO suffer from decomposition to  $E/Z$   $(i\text{PrO})_2(\text{O})\text{PCF}=\text{CFP}(\text{O})(i\text{PrO})_2$  and  $\text{HCF}_2\text{P}(\text{O})(\text{OiPr})_2$ , as was previously reported by Burton.<sup>[9a]</sup> In our studies, the formation of  $(i\text{PrO})_2(\text{O})\text{PCF}_2\text{CF}_2\text{P}(\text{O})(i\text{PrO})_2$ , as caused by a radical decomposition mechanism, was not observed.<sup>[7e]</sup> The analysis of the  $^{19}\text{F}$  NMR spectra of the DMSO solution shows only traces of decomposition products after 25 days of storage.  $^{19}\text{F}$  NMR spectra in DMF reveal a very slow

decomposition. In acetonitrile the copper compound shows a different behavior as far as the stability is concerned. Analysis of the  $^{19}\text{F}$  NMR spectra over 25 days at room temperature reveals that the decrease of the signal intensity follows a first-order decay, with a half-life of  $133\text{ h}$  ( $5.5\text{ d}$ ). This value was calculated by a fit of a first-order exponential decay for the decrease of the  $^{19}\text{F}$  NMR signal integral values (Figure 4). The good agreement of the data with the model of a first-order decay implies a rate-determining step of the decay proceeding over one distinct pathway, presumably a carbenoid-like structure.<sup>[7c,9a]</sup>

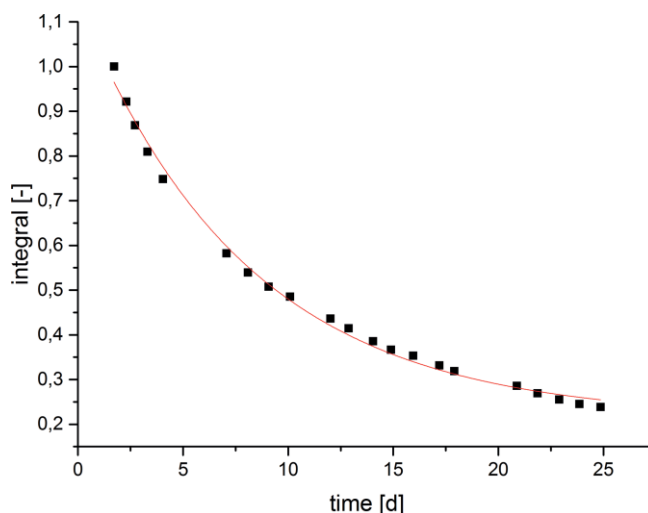


Figure 4. Plot of the signal integral of the  $^{19}\text{F}$  NMR spectra assigned to the organocuprate in acetonitrile monitored for 25 d. The data was fitted with a first-order exponential decay function.

Reactivity studies of the isolated compounds with allyl bromides were carried out in DMF at room temperature (Table 1). Whereas  $\text{BrZnCF}_2\text{P}(\text{O})(\text{OiPr})_2$  (**2**) shows no reaction towards an excess of allyl bromide or methyl 4-bromocrotonate after 2 and 24 hours of reaction time, organocuprate **1** reveals complete conversion within two hours. When  $\text{Cu}^{\text{I}}\text{Br}$  is added to

Table 1. Reactivity studies of isolated  $[\text{Zn}\{\text{Cu}(\text{Br})\{\text{CF}_2\text{P}(\text{O})(\text{OiPr})_2\}_2\}(\text{dmf})_4]$  (**1**) and isolated  $[\text{Zn}(\text{Br})\{\text{CF}_2\text{P}(\text{O})(\text{OiPr})_2\}(\text{thf})_2]$  (**2**) with an excess of allyl bromides.<sup>[a]</sup>

| Compound | Substrate                                                 | Catalyst                   | Time | Yield [%] <sup>[b]</sup> |
|----------|-----------------------------------------------------------|----------------------------|------|--------------------------|
| <b>1</b> | $\text{BrCH}_2\text{CH}=\text{CH}_2$                      | –                          | 2 h  | > 97 %                   |
| <b>1</b> | $\text{BrCH}_2\text{CH}=\text{CHC}(\text{O})\text{OCH}_3$ | –                          | 2 h  | > 97 %                   |
| <b>2</b> | $\text{BrCH}_2\text{CH}=\text{CH}_2$                      | –                          | 2 h  | 0 %                      |
| <b>2</b> | $\text{BrCH}_2\text{CH}=\text{CH}_2$                      | –                          | 24 h | 0 %                      |
| <b>2</b> | $\text{BrCH}_2\text{CH}=\text{CHC}(\text{O})\text{OCH}_3$ | –                          | 2 h  | 0 %                      |
| <b>2</b> | $\text{BrCH}_2\text{CH}=\text{CHC}(\text{O})\text{OCH}_3$ | –                          | 24 h | 0 %                      |
| <b>2</b> | $\text{BrCH}_2\text{CH}=\text{CH}_2$                      | $\text{CuBr}^{\text{[c]}}$ | 2 h  | < 1 %                    |
| <b>2</b> | $\text{BrCH}_2\text{CH}=\text{CH}_2$                      | $\text{CuBr}^{\text{[c]}}$ | 24 h | 14 %                     |
| <b>2</b> | $\text{BrCH}_2\text{CH}=\text{CHC}(\text{O})\text{OCH}_3$ | $\text{CuBr}^{\text{[c]}}$ | 2 h  | 11 %                     |
| <b>2</b> | $\text{BrCH}_2\text{CH}=\text{CHC}(\text{O})\text{OCH}_3$ | $\text{CuBr}^{\text{[c]}}$ | 24 h | 64 %                     |
| <b>2</b> | $\text{BrCH}_2\text{CH}=\text{CH}_2$                      | <b>1</b> <sup>[d]</sup>    | 2 h  | < 1 %                    |
| <b>2</b> | $\text{BrCH}_2\text{CH}=\text{CH}_2$                      | <b>1</b> <sup>[d]</sup>    | 24 h | 5 %                      |
| <b>2</b> | $\text{BrCH}_2\text{CH}=\text{CHC}(\text{O})\text{OCH}_3$ | <b>1</b> <sup>[d]</sup>    | 2 h  | 4 %                      |
| <b>2</b> | $\text{BrCH}_2\text{CH}=\text{CHC}(\text{O})\text{OCH}_3$ | <b>1</b> <sup>[d]</sup>    | 24 h | 24 %                     |

[a] All reactions were carried out in DMF at room temperature. [b] Determined by integration of the  $^{19}\text{F}$  NMR resonances. [c] 3.5 mol-%. [d] 1.6 mol-%.

a solution of  $\text{BrZnCF}_2\text{P}(\text{O})(\text{OiPr})_2$  (**2**) the reaction proceeds as expected, but significantly slower than a stoichiometric amount of organocuprate **1**. A reaction also occurred when catalytic amounts of the isolated cuprate **1** were added to a solution of  $\text{BrZnCF}_2\text{P}(\text{O})(\text{OiPr})_2$  (**2**). On the basis of these observations we conclude that the formation of the organocuprate **1** from  $\text{BrZnCF}_2\text{P}(\text{O})(\text{OiPr})_2$  (**2**) is crucial for the coupling of the  $\text{CF}_2\text{P}(\text{O})(\text{OR})_2$  unit to organic electrophiles. When a catalytic amount of  $\text{Cu}^{\text{I}}$  is added to a solution of  $\text{BrZnCF}_2\text{P}(\text{O})(\text{OiPr})_2$  (**2**) cuprate **1** is formed and can undergo further reactions. It is noteworthy that in our studies  $\text{Pd}^0$  and  $\text{Pd}^{\text{II}}$  show no catalytic activity in Negishi type coupling reactions of  $\text{BrZnCF}_2\text{P}(\text{O})(\text{OiPr})_2$  with allyl halides.

## Conclusions

We isolated  $[\text{Zn}(\text{Br})\{\text{CF}_2\text{P}(\text{O})(\text{OiPr})_2\}(\text{thf})]_2$  and  $[\text{Zn}\{\text{Cu}(\text{Br})\{\text{CF}_2\text{P}(\text{O})(\text{OiPr})_2\}_2(\text{dmf})_4\}]$  for the first time and studied their solid-state molecular structure and spectroscopic properties. Reactivity experiments on the isolated compounds indicate that the formation of the reactive  $[\text{Cu}(\text{Br})\{\text{CF}_2\text{P}(\text{O})(\text{OiPr})_2\}]^-$  is crucial to introduce the  $\text{CF}_2\text{P}(\text{O})(\text{OR})_2$  unit into organic molecules. Stability studies reveal that the choice of solvent is important to increase the stability of the reactive cuprate at room temperature to avoid decomposition.

## Experimental Section

The starting material  $\text{BrCF}_2\text{P}(\text{O})(\text{OiPr})_2$ <sup>[5]</sup> was synthesized as described in the literature. All other chemicals were obtained from commercial sources and used without further purification. Standard high-vacuum techniques were employed throughout all experiments. Nonvolatile compounds were handled in a dry  $\text{N}_2$  atmosphere using Schlenk techniques. IR spectra were recorded with a Bruker Alpha FTIR spectrometer (Bruker Daltonik GmbH, Bremen, Germany) equipped with an ATR unit with a diamond crystal for liquids and solids. NMR spectra were recorded with a Bruker Model Avance III 300 spectrometer ( $^3\text{P}$  121.5 MHz;  $^{19}\text{F}$  282.4 MHz;  $^{13}\text{C}$  75.5 MHz;  $^1\text{H}$  300.1 MHz), positive shifts being downfield from the external standards [TMS ( $^{13}\text{C}$ ;  $^1\text{H}$ ),  $\text{CCl}_3\text{F}$  ( $^{19}\text{F}$ ); 85 %  $\text{H}_3\text{PO}_4$  ( $^3\text{P}$ )].

**$[\text{Zn}(\text{Br})\{\text{CF}_2\text{P}(\text{O})(\text{OiPr})_2\}(\text{thf})]_2$ :** Bromodifluoromethyldiisopropylphosphonate (10.0 g; 34.0 mmol) was slowly added to a slurry of zinc dust (2.26 g; 34.7 mmol) in dry THF (50 mL). After heating the mixture for 5 min at 66 °C, the slurry was stirred at room temperature for 5 h. After filtration the crude product was crystallized from the THF solution at –28 °C. Recrystallization from THF (15 mL) yielded 5.79 g (13.5 mmol, 40 %) of  $[\text{Zn}(\text{Br})\{\text{CF}_2\text{P}(\text{O})(\text{OiPr})_2\}(\text{thf})]_2$  as colorless needles.  $^1\text{H}$  NMR ( $[\text{D}_6]\text{DMSO}$ , room temp.):  $\delta$  = 1.23 (d,  $^3J_{\text{H,H}}$  = 6 Hz, 6 H,  $\text{CH}_3$ ), 1.24 (d,  $^3J_{\text{H,H}}$  = 6 Hz, 6 H,  $\text{CH}_3$ ), 1.74 (m, 4 H,  $\text{OCH}_2$ ), 3.58 (m, 4 H,  $\text{CH}_2$ ), 4.60 (m, 2 H, CH) ppm;  $^{13}\text{C}\{^1\text{H}\}$  NMR ( $[\text{D}_6]\text{DMSO}$ , r.t.):  $\delta$  = 24.2 (d,  $^3J_{\text{P,C}}$  = 5 Hz,  $\text{CH}_3$ ), 24.5 (d,  $^3J_{\text{P,C}}$  = 3 Hz,  $\text{CH}_3$ ), 25.6 (s,  $\text{CH}_2$ ), 67.4 (s,  $\text{OCH}_2$ ), 70.6 (d,  $^2J_{\text{P,C}}$  = 7 Hz, CH), 142.3 (dt,  $^1J_{\text{P,C}}$  = 122,  $^1J_{\text{F,C}}$  = 290 Hz,  $\text{CF}_2$ ) ppm.  $^{19}\text{F}$  NMR ( $[\text{D}_6]\text{DMSO}$ , r.t.):  $\delta$  = –124.7 (d,  $^2J_{\text{P,F}}$  = 85 Hz, 2 F) ppm;  $^3\text{P}$  NMR ( $[\text{D}_6]\text{DMSO}$ , r.t.):  $\delta$  = 12.7 (tt,  $^2J_{\text{P,F}}$  = 85,  $^3J_{\text{P,H}}$  = 7 Hz, 1 P) ppm. IR (ATR):  $\tilde{\nu}$  = 2982 (w), 2936 (w), 2898 (w), 2879 (w), 1465 (w), 1452 (w), 1387 (w), 1377 (w), 1345 (w), 1199 (m), 1179 (m), 1143 (w), 1099 (w), 1072 (s), 1015 (s), 990 (m), 931 (m), 898 (m), 872 (m), 759 (w), 739 (w), 676 (m), 618 (w, shoulder), 528 (m), 407 (w), 390 (w)  $\text{cm}^{-1}$ .

**$[\text{Zn}\{\text{Cu}(\text{Br})\{\text{CF}_2\text{P}(\text{O})(\text{OiPr})_2\}_2(\text{dmf})_4\}]$ :** A sample of  $[\text{Zn}(\text{Br})\{\text{CF}_2\text{P}(\text{O})(\text{OiPr})_2\}(\text{thf})]_2$  (2.30 g; 5.37 mmol) was dissolved in dry DMF (10 mL), and  $\text{CuBr}$  (770 mg; 5.43 mmol) was added. The mixture was stirred at room temperature for 2 h. Dimethylformamide was evaporated and THF (15 mL) was added to the residue. After filtration the product was crystallized at –28 °C from the THF solution to yield 937 mg (0.91 mmol; 34 %) of  $[\text{Zn}\{\text{Cu}(\text{Br})\{\text{CF}_2\text{P}(\text{O})(\text{OiPr})_2\}_2(\text{dmf})_4\}]$  as colorless needles.  $^1\text{H}$  NMR ( $[\text{D}_6]\text{DMSO}$ , room temp.):  $\delta$  = 1.23 (d,  $^3J_{\text{H,H}}$  = 6 Hz, 12 H,  $\text{CH}_3$ ), 1.24 (d,  $^3J_{\text{H,H}}$  = 6 Hz, 12 H,  $\text{CH}_3$ ), 2.73 (s, 12 H,  $\text{CH}_3$ ), 2.90 (s, 12 H,  $\text{CH}_3$ ), 4.62 (m, 4 H, CH), 7.96 (s, 4 H, O=CH) ppm.  $^{13}\text{C}\{^1\text{H}\}$  NMR ( $[\text{D}_6]\text{DMSO}$ , r.t.):  $\delta$  = 24.04 (d,  $^3J_{\text{P,C}}$  = 5 Hz,  $\text{CH}_3$ ), 24.10 (d,  $^3J_{\text{P,C}}$  = 3 Hz,  $\text{CH}_3$ ), 30.8 (s,  $\text{NCH}_3$ ), 35.8 (s,  $\text{NCH}_3$ ), 69.5 (d,  $^2J_{\text{P,C}}$  = 7 Hz, CH), 162.3 (s, O=CH) ppm.  $^{13}\text{C}\{^{19}\text{F}\}$  dept45 NMR ( $[\text{D}_6]\text{DMSO}$ , r.t.):  $\delta$  = 149.3 (d,  $^1J_{\text{P,C}}$  = 114 Hz) ppm.  $^{19}\text{F}$  NMR ( $[\text{D}_6]\text{DMSO}$ , r.t.):  $\delta$  = –114.0 (d,  $^2J_{\text{P,F}}$  = 91 Hz, 2 F) ppm.  $^3\text{P}$  NMR ( $[\text{D}_6]\text{DMSO}$ , r.t.):  $\delta$  = 13.7 (tt,  $^2J_{\text{P,F}}$  = 91,  $^3J_{\text{P,H}}$  = 7 Hz, 1 P) ppm. IR (ATR):  $\tilde{\nu}$  = 2976 (w), 2931 (w), 2878 (w), 2811 (w), 1643 (s), 1495 (w), 1435 (w), 1419 (w), 1376 (m), 1252 (w), 1221 (m), 1178 (w), 1143 (w), 1109 (m), 1053 (m), 1002 (m), 981 (m), 901 (w), 872 (w), 766 (w), 744 (w), 682 (m), 608 (w), 519 (w), 480 (w, shoulder)  $\text{cm}^{-1}$ .

**Stability Measurements:** Three Young NMR tubes were each charged with a sample of  $[\text{Zn}\{\text{Cu}(\text{Br})\{\text{CF}_2\text{P}(\text{O})(\text{OiPr})_2\}_2(\text{dmf})_4\}]$ . The samples were dissolved in DMF, DMSO, or acetonitrile.  $^{19}\text{F}$  NMR spectra were recorded periodically over one month. The samples were stored at room temperature exposed to daylight. The spectra of the sample in acetonitrile were analyzed with Mestrenova. The decrease of the signal integrals was fitted with a first-order exponential decay.

**General Procedure for the Reactivity Studies of  $[\text{Zn}\{\text{Cu}(\text{Br})\{\text{CF}_2\text{P}(\text{O})(\text{OiPr})_2\}_2(\text{dmf})_4\}]$ :** A solution of  $[\text{Zn}\{\text{Cu}(\text{Br})\{\text{CF}_2\text{P}(\text{O})(\text{OiPr})_2\}_2(\text{dmf})_4\}]$  (212 mg) in DMF (10 mL) was charged with an excess of the substrate. The solution was stirred for 2 h at room temperature.

**General Procedure for the Reactivity Studies of  $[\text{Zn}(\text{Br})\{\text{CF}_2\text{P}(\text{O})(\text{OiPr})_2\}(\text{thf})]_2$ :** A solution of  $[\text{Zn}(\text{Br})\{\text{CF}_2\text{P}(\text{O})(\text{OiPr})_2\}(\text{thf})]_2$  (607 mg) in DMF (5 mL) was charged with an excess of the substrate. The catalyst was added to the solution and stirred for 24 h at room temperature.

CCDC 1547399 (for  $[\text{Zn}(\text{Br})\{\text{CF}_2\text{P}(\text{O})(\text{OiPr})_2\}(\text{thf})]_2$ ) and 1547400 (for  $[\text{Zn}\{\text{Cu}(\text{Br})\{\text{CF}_2\text{P}(\text{O})(\text{OiPr})_2\}_2(\text{dmf})_4\}]$ ) contain the supplementary crystallographic data for this paper. These data can be obtained free of charge from The Cambridge Crystallographic Data Centre.

## Acknowledgments

This work was supported by Merck KGaA (Darmstadt, Germany). Solvay (Hannover, Germany) is gratefully acknowledged for donating chemicals. We also acknowledge support by the Deutsche Forschungsgemeinschaft (Core Facility GED@BI, Mi477/21-1). We thank Dr. Julia Bader and Markus Wiesemann for helpful discussions.

**Keywords:** Fluorine · Halogens · Zinc · Phosphorus · Cuprates

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Received: May 10, 2017

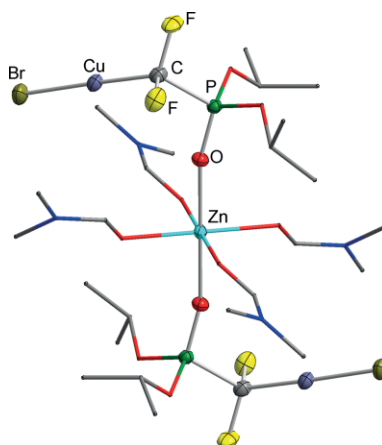


## Reactive Phosphonate Species

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**Solid-State Structure of a Bromo-(difluoromethylenephosphonato)-cuprate – a Key Intermediate for the Synthesis of  $\alpha,\alpha$ -Difluoromethylene-phosphonates**



Many approaches to introduce the  $\text{CF}_2\text{P}(\text{O})(\text{OR})_2$  unit into organic molecules make use of  $\text{Cu}^{\text{I}}$ -catalyzed couplings of  $\text{BrZnCF}_2\text{P}(\text{O})(\text{OR})_2$ . The constitution and molecular structures of the organozinc precursor and the catalytically relevant copper/zinc complex are still unknown. Here we report on the isolation of both species and the elucidation of their structure by single-crystal X-ray diffraction.

DOI: 10.1002/ejic.201700526